Group Project - Final Paper

Medicaid Group 2
Annie Qin, Marcello Borromeo, Dhruv Nair, Mo Mia

Arizona State University
W. P. Carey School of Business

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Dr. Tam McCreless

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Introduction to Medicaid



Medicaid is a federal–state entitlement program that finances healthcare for millions of low-income Americans, and a growing proportion of its budget is dedicated to prescription drugs. Pharmaceutical spending now exceeds $65 billion annually and continues to rise due to the expansion of chronic-disease therapies, the introduction of specialty biologics, and persistent utilization of high-cost brand medications, even when generics exist. Understanding where Medicaid spends the most money, and where savings are feasible without compromising access, is essential for policymakers. This project analyzes national Medicaid drug utilization data to address five central questions: which drugs account for the largest share of spending (Q1), which therapeutic classes account for the largest share of spending (Q2), where brand-to-generic substitution could result in measurable savings (Q3), which drugs have therapeutically equivalent alternatives (Q4), and what underlying factors, including regional patterns, drive these costs (Q5). Together, these questions frame a comprehensive look at cost concentration, substitution opportunities, and geographic variability in Medicaid's pharmaceutical landscape.

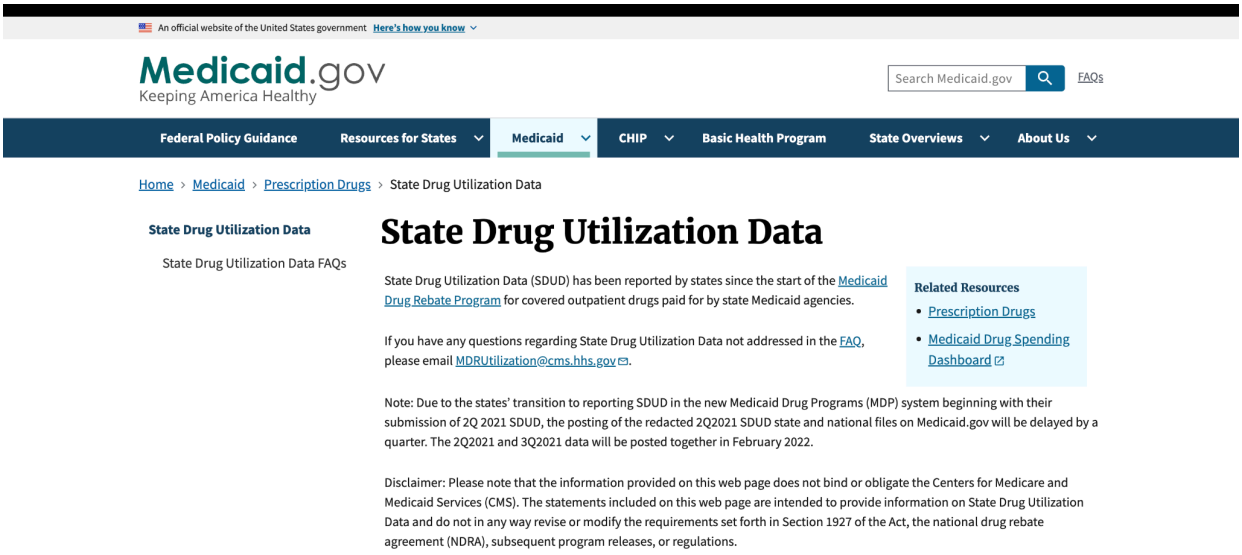
Problem / Issue Addressed



The first issue addressed in this project involves identifying the drugs and drug classes that constitute the largest portion of Medicaid pharmaceutical spending. Medicaid's limited resources make it particularly important to understand which products impose the greatest financial burden, as this helps highlight where targeted intervention could be most effective. The second issue concerns determining whether there are realistic opportunities for Medicaid to reduce spending by encouraging the use of lower-cost generic equivalents; this requires both measuring the price difference between brand and generic formulations and quantifying how often brand versions are still used when generics exist. The third issue addresses which drugs actually have therapeutic substitutes and whether substitution is clinically feasible, since some drugs, particularly in oncology and certain autoimmune conditions, have no generic or biosimilar alternatives. Finally, the project explores the drivers of drug costs, including whether differences in price per unit are attributable to brand status, drug class, or regional variation across Medicaid programs nationwide. By answering these questions collectively, the analysis aims to reveal not only the structure of Medicaid spending but also the most actionable opportunities for policy change.

Steps to Solve the Problem: Data Sources

To



To address these questions, the analysis draws on data from the National Drug Code Directory and the 2024 All States Drug Utilization dataset, which together provide a comprehensive picture of utilization, reimbursement, and drug characteristics across all states. These datasets include National Drug Codes (NDCs), nonproprietary drug names, brand/generic identifiers, pharmacologic class labels, drug class assignments, units reimbursed, total Medicaid amount reimbursed, and cost per unit.

To ensure all of this information was easily visible in one place, the National Drug Code Directory dataset and the 2024 All States Drug Utilization dataset needed to be joined. The common key between the datasets was the NDC column. However, the format of the NDC in the datasets differed from each other when downloaded in their raw state. To standardize the NDCs into the 5 labeler - 4 product code format, the 2024 All States File only needed to take the 9 leftmost digits. However, with the National Drug Code Directory, it was already split into the labeler code and the product code, but both codes had the wrong number of digits. After splitting the code at the hyphen, we added one more leading 0 to fix the labeler code into 5 digits, and ensured the product code was in a 4 digit format, only adding

another leading 0 if it was not there. Then we would concatenate the labeler and product code once the digits in each were fixed, to then have the correct 9 digit NDC. Now, with both datasets in the correct NDC format, they were inner joined to create the master dataset.

To answer the project questions, new columns were also added for the analysis. For Q5, information on the region was required. By using the 'State' column, we used an 'If' function in Excel to generate what region (South, West, Northeast) of the United States the state was in. Additionally, to answer other questions, 'Cost per Unit' column was created, by dividing the 'Medicaid Amount Reimbursed' by the 'Units Reimbursed' column. For questions 3 and 4, the analysis required knowing if the drug was a brand drug or a generic drug. To classify this, we used an Excel function to see if the proprietary name and the non-proprietary name columns were the same. If they were, then it was a generic drug; if they were different, it was a brand drug. Finally, some of the questions required knowing the drug class. After attending office hours, Professor Tam recommended using ChatGPT to read through the pharm_classes column and have AI classify the drug class based on that column. ChatGPT was able to create the column and give the drug class for each row of data.

Finally, after cleaning missing values and removing the ambiguous class labeled "Other," the dataset contained 195,003 usable records covering the breadth of Medicaid outpatient drug utilization. This dataset provided the necessary detail to examine pricing disparities, model substitution savings, and characterize regional spending differences.

Steps to Solve the Problem: Analytical Techniques

The analytical approach for Q1 and Q2 relied on aggregating Medicaid Amount Reimbursed to identify the highest-spend drugs and highest-spend therapeutic classes. Pivot tables, grouped summaries, and descriptive statistics illustrated the degree to which spending is concentrated among chronic-disease treatments and expensive specialty medications. For Q3, which focused on savings potential, the analysis isolated 537 drugs that have both brand and generic versions. Because outliers in drug pricing heavily skew mean values, median cost per unit was computed for brand and generic formulations separately, and weighted-average generic prices were calculated to provide a realistic estimate of typical generic spending. Per-unit savings were then multiplied by the total number of brand units reimbursed nationally to estimate total potential savings. Substitution rates were modeled from 50% to 100% to represent feasible policy scenarios, and an additional class-adjusted scenario incorporated realistic substitution likelihoods for each therapeutic category.

Questions 4 and 5 required more detailed modeling. To examine drugs with equivalent alternatives and to understand cost drivers, categorical data—such as drug class, state, and region—were encoded numerically and combined with usage variables in a Random Forest regression model predicting log-transformed cost per unit. This method was chosen because linear models performed poorly in the presence of skewed cost distributions and multicollinearity among reimbursement variables. Finally, to evaluate regional variation, the analysis summarized costs by region, compared median and mean cost per unit, conducted Kruskal–Wallis tests to determine whether observed differences were statistically significant, and calculated the top drugs and top drug classes contributing to spending in each region. Together, these techniques provided a multi-layered understanding of how spending, prices, and substitution potential vary across drug types and geographic areas.

How Q3 was solved

We'll take a moment to show you how Q3, Opportunities for Savings, was solved. We believe this was the most technical and important part of this project since we are trying to find feasible ways to reduce costs for Medicaid.

To estimate Medicaid's true opportunities for savings, we first identified which drugs could realistically be substituted—that is, which drugs had both brand and generic versions available. We did this by grouping the entire dataset by nonproprietary name and checking whether each ingredient appeared in both the "Brand" and "Generic" categories. Only drugs with at least one brand NDC and one generic NDC were included. This filtering step gave us 537 eligible drugs, which represent every realistic substitution opportunity in the dataset.

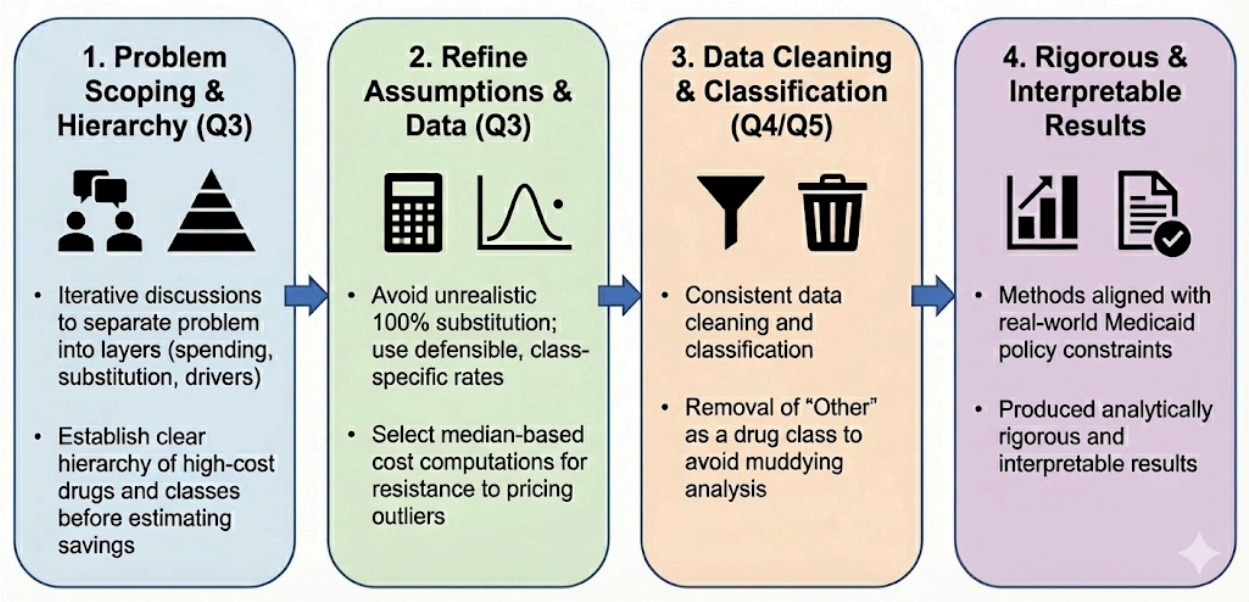
For each of these 537 drugs, we calculated two measures of generic pricing: the median generic cost per unit and a weighted-average generic cost per unit based on utilization. Using both allowed us to compare a clean, outlier-resistant estimate (median) with a more realistic market estimate (weighted). We then compared these values with the median brand cost per unit to compute the savings per unit—how much Medicaid could save every time a brand unit is replaced with its generic equivalent.

Next, we scaled these per-unit savings by multiplying them by the total number of brand units reimbursed for each drug. This step converts price differences into real-world dollars and identifies which drugs generate the largest savings impact at Medicaid's current utilization levels.

To make the model realistically actionable, we then incorporated substitution rates by drug class, since some therapeutic areas substitute more easily than others. For example, Mental Health drugs tend to have higher substitution potential, while Oncology drugs have very low substitutability. This allowed us to compute a class-adjusted "realistic savings" estimate, not just a theoretical maximum.

Finally, we ranked drugs by their total potential and realistic savings, plotted the brand vs. generic cost differences, and created a substitution savings curve to show how total savings grow from 10% to 100% substitution. This analysis revealed that even modest increases in generic use can unlock significant value, and that most of the savings come from a small set of high-cost, high-volume brand drugs.

Steps to Solve the Problem: Scoping and Office Hour Discussions



Our analytical scope was refined through iterative discussions, which emphasized separating the broader problem into investigative layers: determining where spending is concentrated, assessing substitution feasibility, and exploring systemic drivers of cost. Guidance showed us the importance of establishing a clear hierarchy of high-cost drugs and classes before estimating savings. In Q3, our discussions stressed avoiding unrealistic assumptions, like 100% substitution across all drugs, and instead incorporating more defensible substitution rates like 50 to 75%. Median-based cost computations were selected over means due to their resistance to extreme pricing outliers, which were especially relevant in categories like oncology and autoimmune biologics. For Q4 and Q5, we wanted to retain completeness in data cleaning and classification, so we removed "Other" as a drug class to avoid muddying the analysis (and also because this class is uninterpretable). These scoping steps helped ensure that our methods aligned with real-world Medicaid policy constraints and produced results that are analytically rigorous and interpretable.

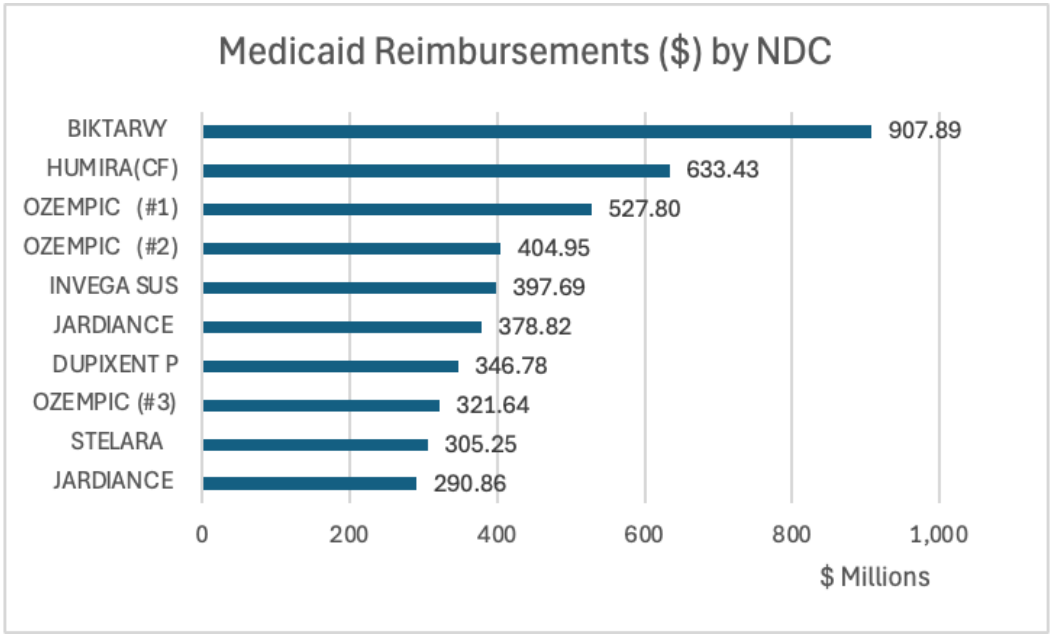
Challenges Faced & Resolutions

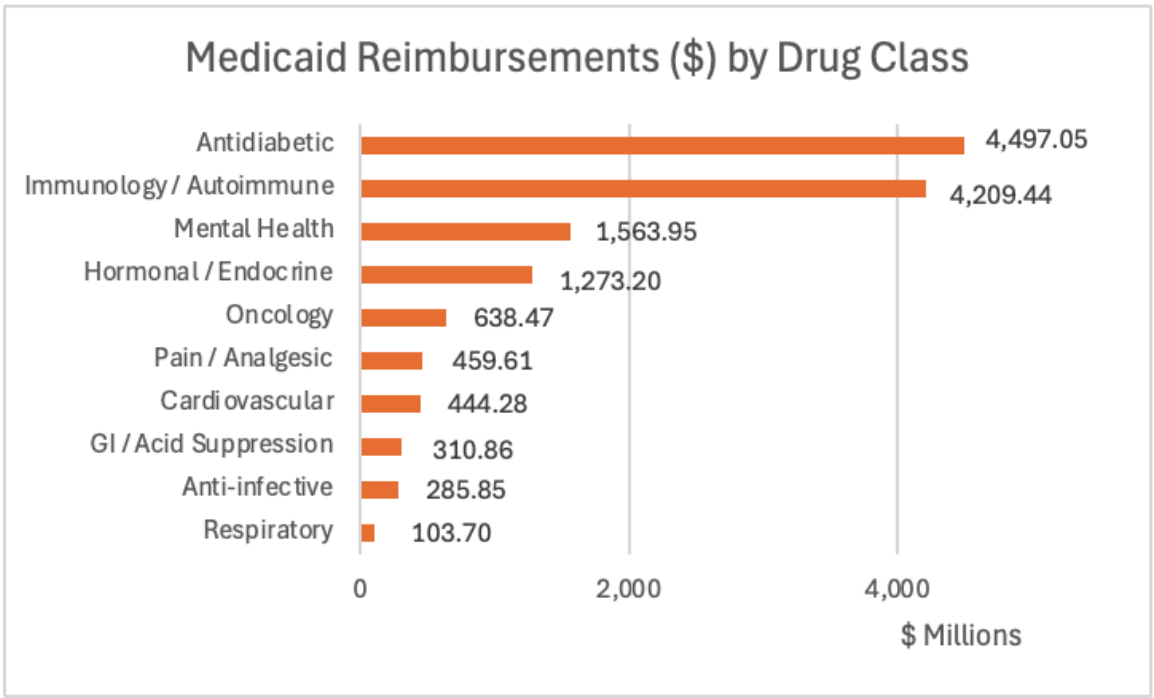
NDC CODE	Stat	Regio	Quart	Product Nam	Units Reimburse	Medicaid Amount Reimburse	Cost per Un	PROPRIETARYNAME	NONPROPRIETARYNAME	BRAND OR GENERIC	LABELERNAM	PHARM_CLASSES	Drug Class
000020152	HI	West	4	ZEPBOUND 2			#DIV/0!	Zepbound	tirzepatide	Brand	Eli Lilly and Compa	G-Protein-linked Rec	Antidiabetic
000020213	AK	West	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AL	South	4	HUMULIN R	55.6	144.56	2.6	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AL	South	3	HUMULIN R	39.75	103.35	2.6	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AL	South	2	HUMULIN R	102.3	625.23	6.11	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AL	South	1	HUMULIN R	165.65	1091.01	6.59	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	4	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	3	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	3	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AR	South	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AZ	West	3	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	AZ	West	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	4	HUMULIN R	394.8	2168.57	5.49	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	4	HUMULIN R	178.9	454	2.54	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	3	HUMULIN R	816.25	4875.59	5.97	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	3	HUMULIN R	129.4	281.3	2.17	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	2	HUMULIN R	849.55	3069.21	3.61	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	2	HUMULIN R	1737.4	11994.93	6.9	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	1	HUMULIN R	2404.6	23275.11	9.68	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CA	West	1	HUMULIN R	591.95	2522.46	4.26	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	4	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	3	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CO	West	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CT	Northeast	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	CT	Northeast	1	HUMULIN R	87	612.42	7.04	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	DC	South	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	DC	South	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	DE	South	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	4	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	4	HUMULIN R	67.7	267.6	3.95	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	3	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	3	HUMULIN R	87.2	361.24	4.14	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	2	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	2	HUMULIN R	140.85	524.67	3.73	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	1	HUMULIN R	199.8	980.11	4.91	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic
000020213	FL	South	1	HUMULIN R			#DIV/0!	Humulin	Insulin human	Brand	Eli Lilly and Compa	Insulin [CS], Insulin [E	Antidiabetic

A significant challenge throughout the analysis was ensuring consistent identification of drugs across thousands of NDC codes, which vary widely in naming conventions and formatting. To resolve this, extensive cleaning standardized drug names, removed formulation descriptors (such as “tablet,” “solution,” or “extended-release”), and grouped NDCs into accurate ingredient-level categories. A second challenge emerged when attempting to use linear regression to estimate generic costs, which resulted in very unrealistic predictions—including negative total savings. This failure stemmed from the highly skewed cost distribution and correlated features such as Units Reimbursed and Medicaid Amount Reimbursed. After removing problematic predictors, the model still behaved erratically, which led us to pivoting toward a gradient boosting decision tree model that better captured nonlinear pricing patterns and produced results consistent with empirical median-based estimates. Furthermore, the dataset required careful handling of missing values, categorical inconsistencies, and region/state alignment issues. Once cleaned, the data allowed for reliable modeling of cost drivers, identification of equivalent drugs, and regional comparisons.

Results: Visualizations & Analysis

Q1–Q2: High-Cost Drugs and Classes



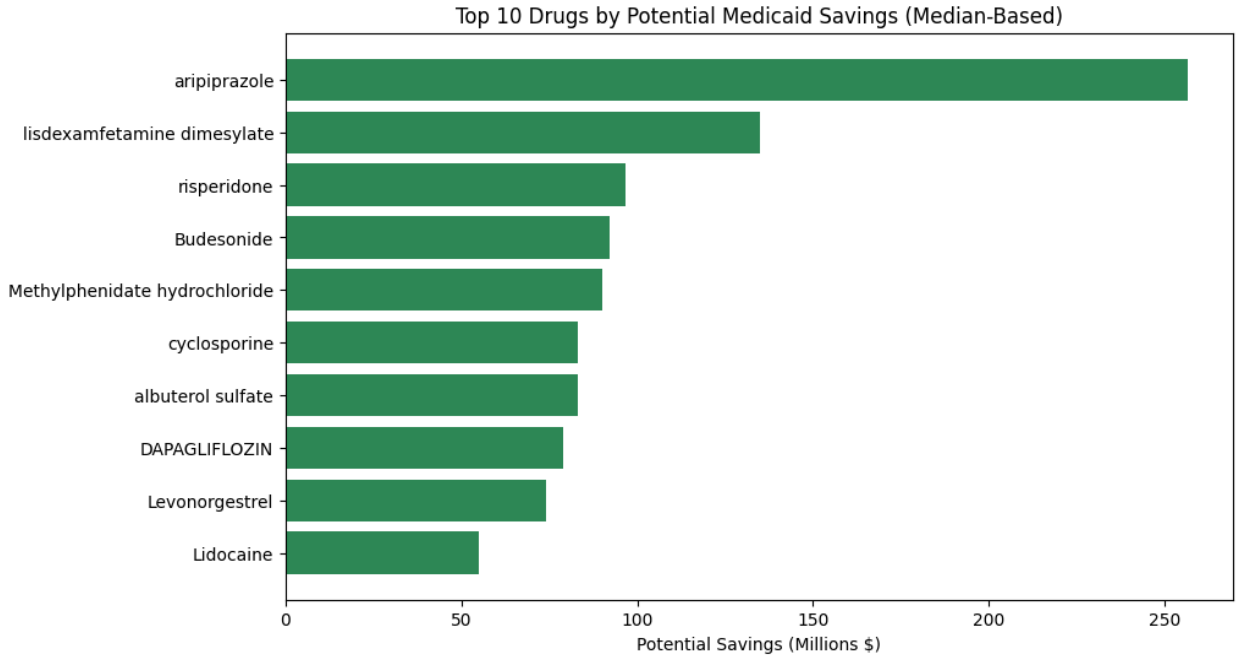


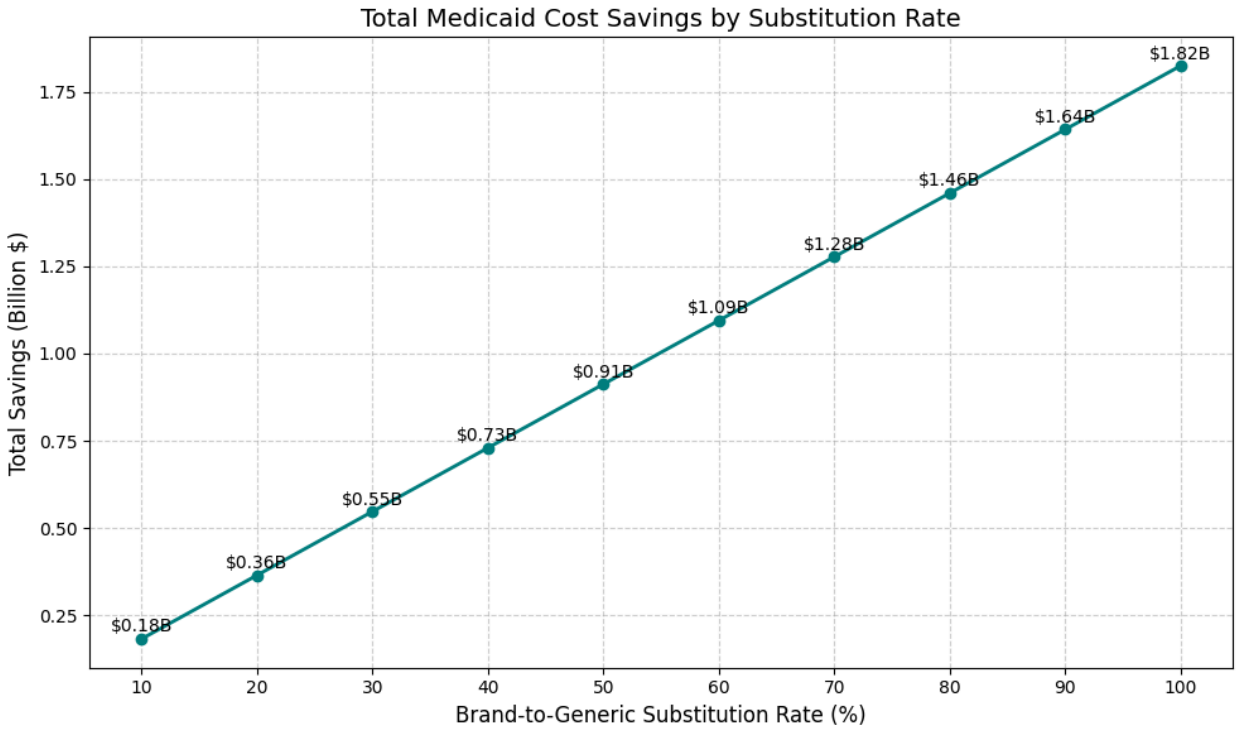
Our results show that Medicaid spending is heavily concentrated in a small number of high-cost brand-name drugs. Just a handful of products—most notably Biktarvy, Humira, and several Ozempic (semaglutide) NDCs—account for hundreds of millions of dollars each in reimbursement, far exceeding the rest of the distribution. This concentration at the drug level indicates that Medicaid's costs are driven not by thousands of moderately priced drugs, but by a narrow set of expensive therapies with large patient populations.

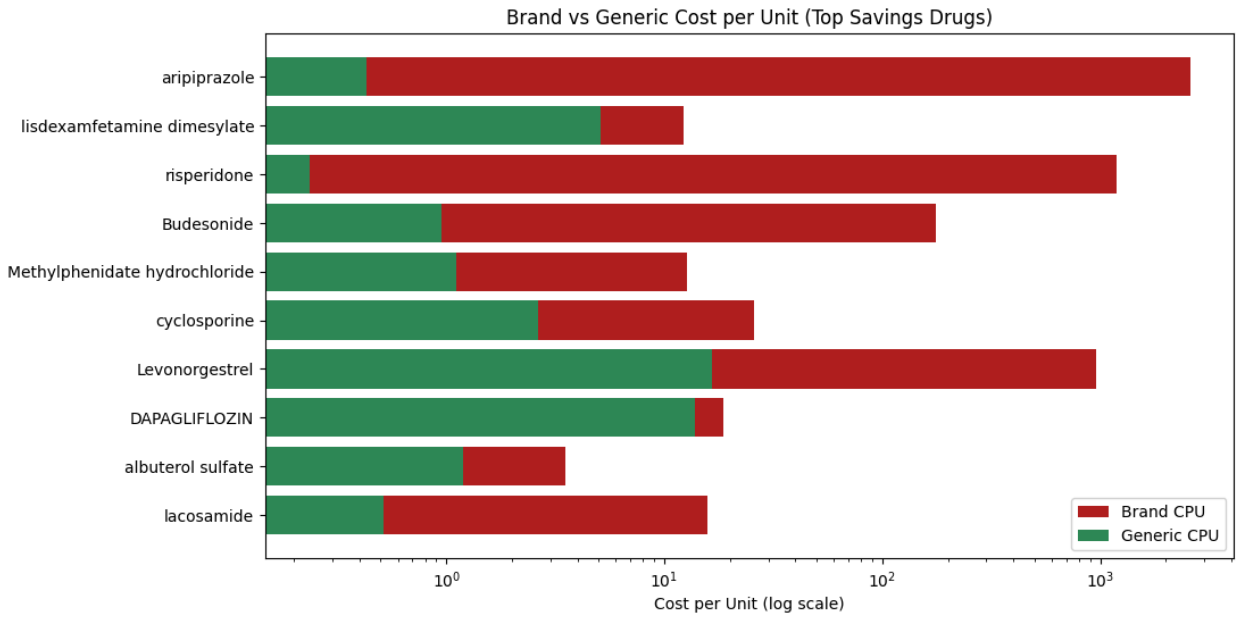
At the class level, spending is similarly concentrated: Antidiabetic and Immunology/Autoimmune drugs dominate national reimbursement totals, together making up roughly 32% of Medicaid's pharmaceutical spending in this dataset. These classes treat long-term chronic illnesses—such as diabetes and autoimmune conditions—which require continuous medication usage and therefore exert sustained pressure on Medicaid budgets.

This pattern tells us that Medicaid's budget challenge is targeted, not systemic! The financial strain comes primarily from a small group of high-cost, high-utilization therapies rather than broad inefficiency across the entire drug market. Recognizing this helps narrow where policy interventions or cost-control efforts can have the greatest impact.

Q3: Opportunities for Savings Through Generic Substitution



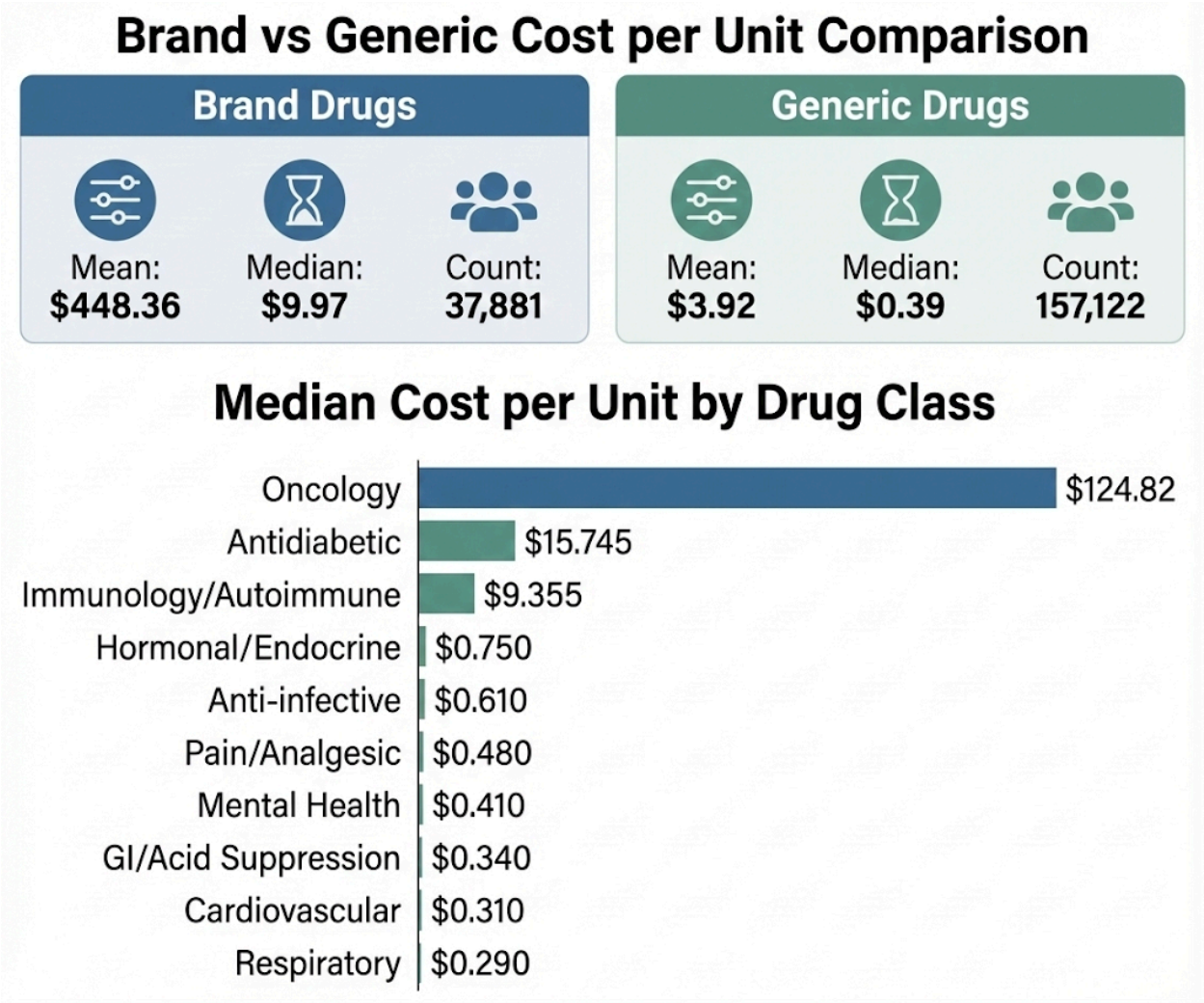




The savings analysis focused on 537 drugs with both brand and generic formulations and found substantial opportunities for cost reduction. Median cost per unit for brand-name drugs was consistently far higher than for their generic equivalents, often by orders of magnitude. After calculating per-unit savings and multiplying them by the total number of brand units reimbursed, the total potential annual savings ranged from approximately $900 million at a 50% substitution rate to $1.8 billion under full substitution. A more realistic scenario that adjusted substitution rates based on therapeutic class—recognizing that oncology and biologics have limited substitutability—estimated annual savings at around $1.2 billion. Importantly, these savings were highly concentrated among a small number of drugs. Aripiprazole, lisdexamfetamine dimesylate, risperidone, budesonide, and albuterol sulfate were among the top contributors, reflecting both high price gaps and substantial brand utilization. The interpretation is clear: Medicaid does NOT need broad, system-wide substitution reform to generate meaningful savings. Instead, targeting a focused set of high-volume brand drugs where generics already exist could capture the majority of the savings with minimal disruption to clinical practice. This concentration also suggests that

provider education, formulary adjustments, and improved monitoring could be sufficient to unlock substantial cost reductions.

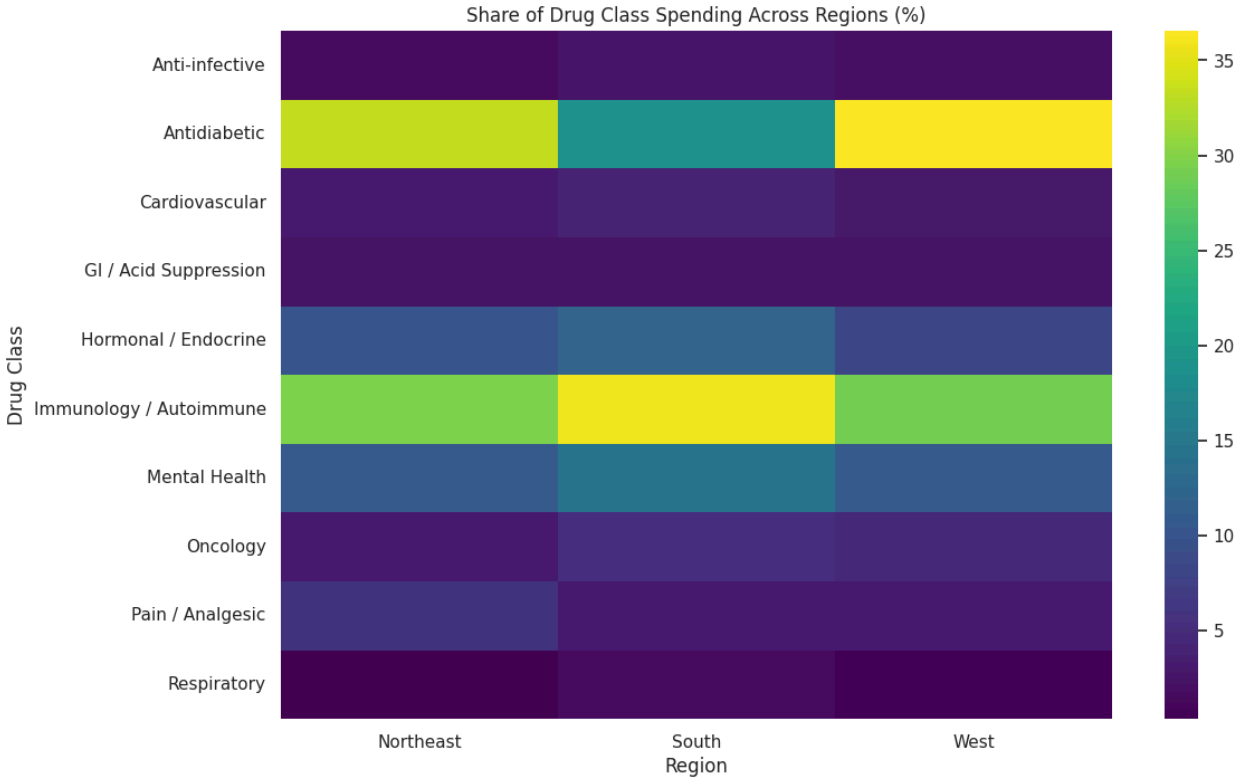
Q4: Drugs with Equivalent Alternatives

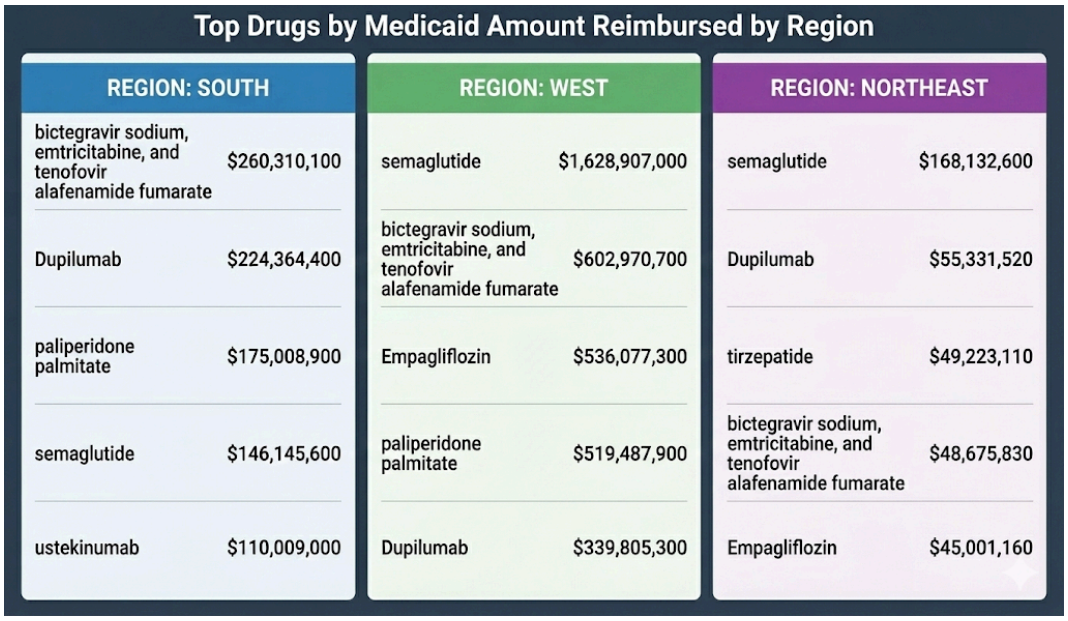


To understand where substitution is feasible, we examined cost differences between brand and generic drugs and identified classes with therapeutically comparable alternatives. Brand drugs exhibited a median cost per unit of $9.97, compared with only $0.39 for generics—a nearly twenty-fivefold difference. The largest price disparities were observed in chronic-disease categories such as Mental Health, Cardiovascular, GI/Acid Suppression, and Pain/Analgesic medications, where generics are widely available and clinically interchangeable. In contrast, Oncology and Immunology/Autoimmune drugs

showed extremely high costs, with median CPUs of $124.82 and $9.36 respectively, reflecting limited generic and biosimilar availability. The Random Forest model confirmed that brand status is the strongest predictor of cost, followed by drug class and ingredient identity, while geographic factors were comparatively minor. This pattern emphasizes that meaningful substitution potential exists primarily in non-specialty chronic-disease categories. The interpretation is that Medicaid should focus substitution initiatives where generic costs are low, availability is high, and therapeutic interchangeability is well established—rather than attempting blanket substitution policies that would be ineffective in specialty drug markets.

Q5: Drivers of Cost and Regional Differences





The regional analysis demonstrated that drug spending varies significantly across the United States. The West region had the highest total spending—nearly $9.9 billion—and also showed the highest mean cost per unit, suggesting deeper reliance on expensive chronic-disease therapies. The South and Northeast regions also displayed distinct spending patterns. A Kruskal–Wallis test confirmed that the differences in cost per unit across regions were statistically significant, indicating that geographic variation is not random. However, despite these differences, all regions relied heavily on the same high-cost drugs, such as semaglutide, HIV antivirals, and autoimmune biologics. This consistency suggests that regional price and spending variation is driven more by differences in utilization volume and patient population health needs rather than geographic differences in drug mix. Heatmap analysis reinforced this conclusion, showing that high-cost therapeutic classes dominate spending across all regions. The interpretation is that while regional pricing and utilization patterns differ, national policy strategies—such as promoting biosimilars, strengthening generic substitution, or improving management of chronic conditions—are likely to yield benefits across all states, since the underlying drivers of cost are broadly consistent nationwide.

Discussion / Implications

Our results collectively show that Medicaid's pharmaceutical spending challenges stem from a combination of high-cost specialty drugs and persistent brand utilization in categories where generics are available. Q1 and Q2 highlight that spending is extremely concentrated, with a small set of drugs and chronic-disease classes accounting for the majority of costs. This concentration suggests that Medicaid can achieve significant impact by focusing on strategic policy interventions rather than across-the-board cuts. Q3 reinforces this by demonstrating that targeted increases in generic substitution could save up to $1.8 billion annually, with realistic savings around $1.2 billion. Importantly, most of these savings come from just a handful of drugs, meaning that highly focused initiatives could yield disproportionate returns. Q4 reveals that the most viable substitution opportunities lie in non-specialty categories, where generics are abundant and therapeutically equivalent, while specialty biologics remain cost-intensive due to limited competition. Finally, Q5 shows that while regional differences exist, they do not alter the fundamental drivers of drug spending; Medicaid programs across the country struggle with the same expensive classes and chronic conditions. Together, these findings suggest a set of actionable policy pathways: promote generic and biosimilar utilization where clinically appropriate, negotiate stronger rebates for high-cost brand drugs, and implement utilization management strategies in chronic-disease categories that dominate Medicaid's budget.

Conclusion

This analysis demonstrates that Medicaid's pharmaceutical spending is shaped by a narrow set of high-cost drugs and specialty therapeutic classes, amplified by continued brand utilization where generics could be used instead. The evidence shows that Medicaid could save between $900 million and $1.8 billion annually through targeted generic substitution alone, with a realistic estimate of around $1.2 billion—most of it driven by a small number of drugs. While regional cost differences exist, the most expensive drugs and therapeutic categories are consistent nationwide, suggesting that national-level policy interventions would be broadly effective. Overall, the results indicate that Medicaid can reduce spending without limiting patient access by implementing focused, data-driven strategies that encourage appropriate substitution, support biosimilar adoption, improve chronic disease management, and concentrate reform efforts on the highest-cost drug categories.

Link to Project Files

Merged Datasets File (and 2 independent datasets) - Excel File

Merged Datasets File (and 2 independent datasets) - Excel File saved onto Google Sheets

Merged Datasets File - Python Code

Q1-Q2 Analysis - Excel File

Q3 Analysis - Colab File

Q4-5 Analysis - Colab File